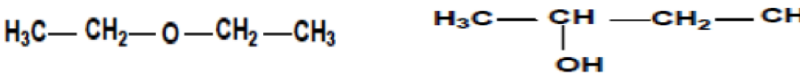
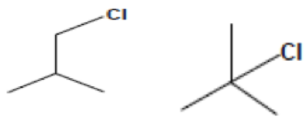
Faculté de Médecine de Constantine
Département de Médecine
1ère Année Médecine

Année Universitaire
2024 / 2025

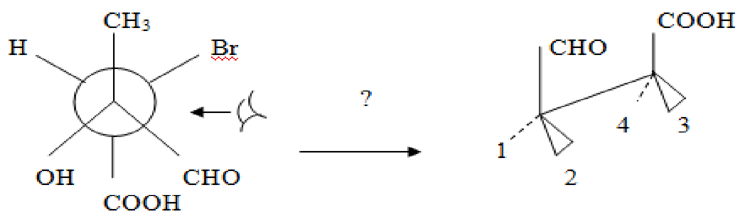
T.D. N° 10 DE CHIMIE

EXERCICE N° 01:
Quelle relation d'isomérie existe-t-il entre chaque paire de molécules ?

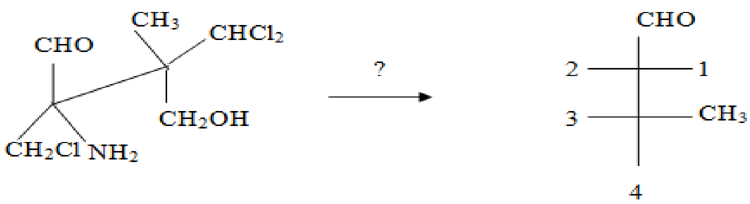




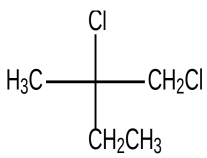
EXERCICE N° 02 :
A- Identifier 1, 2, 3 et 4

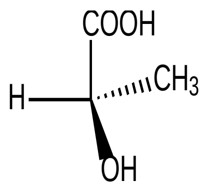


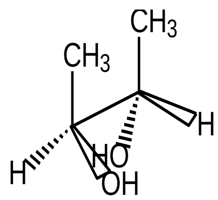
B- Identifier 1, 2, 3 et 4

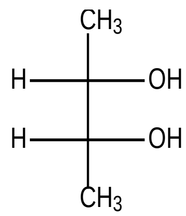


EXERCICE N° 03:
Donner la configuration absolue



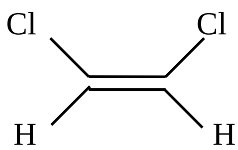






Donner en Cram : (S)-2-amino-2-méthyléthanal

EXERCICE N° 04 :
Donnez la stérioisomérie géométrique Cis-trans ou Z - E des molécules suivantes

a- 

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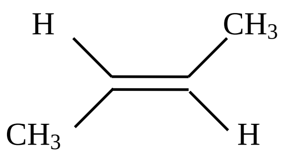
b- 

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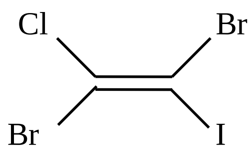
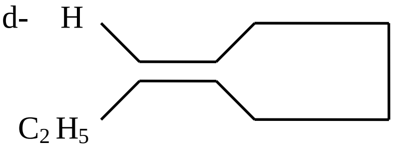
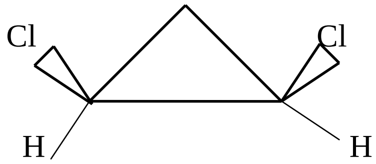
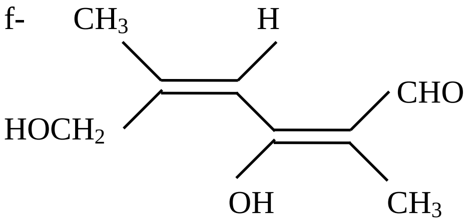
c- 

d- 

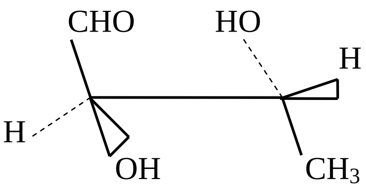
;

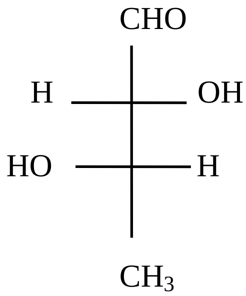
e- 

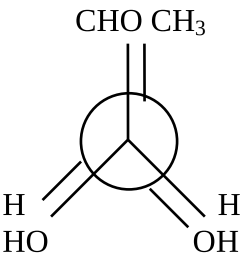
f- 

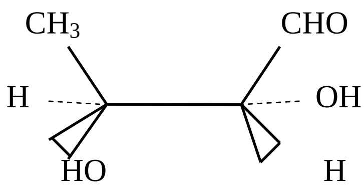
<u>EXERCICE N° 05</u>:

On a les stéréoisomères suivants:

a- 

b- 

c- 

d- 

1- Comparer b et d. Justifier votre réponse.
2- Comparer c et d. Justifier votre réponse.
3- Les molécules a et c sont diastéréoiomères. Expliquer.
4- Donner la configuration relative de la molécule a.

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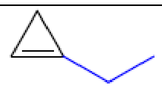
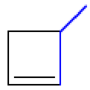
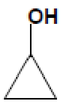
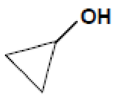
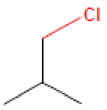
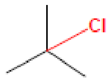
CORRIGE TYPE DU T.D. N° 10 DE CHIMIE

EXERCICE N° 01:

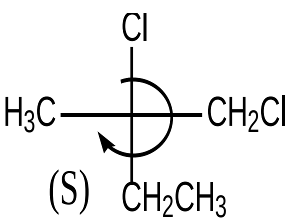
$H_3C-CH_2-O-CH_2-CH_3$	$H_3C-CH(OH)-CH_2-CH_3$	Isomères de fonction
		Isomères de chaîne
		Identiques
		Isomères de position

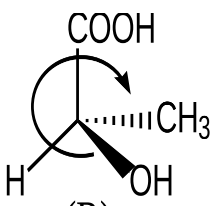
EXERCICE N° 02 :

A- 1 = CH3, 2 = OH, 3 = H, 4 = Br.   B- 1= NH_2, 2= CH_2Cl, 3= $CHCl_2$, 4= CH_2OH.

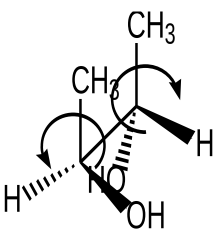
EXERCICE N° 03:

1-

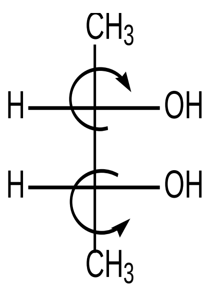




(R)

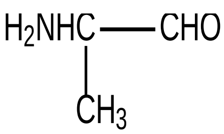


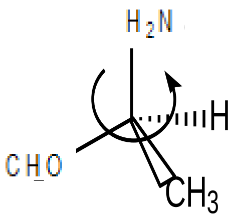
(2S,3S)



(2S, 3R)

2-





(S)

EXERCICE N° 04:

a- Cis ; b- Trans ; c- Z ; d- Rien ; e- Cis ; f- Z , E

EXERCICE N° 05:

1- b et d sont identiques car elles ont la même configuration absolue, soit 2R 3S.

2- c et d sont énantiomères car elles ont des configurations absolues inversées. 2S 3R pour c et 2R 3S pour d.

3- a et c sont diastéréoisomères car un seul carbone est inversé. 2R 3R pour a et 2S 3R pour c.

4- a est de configuration relative érythro car les substituants de même priorité sont éclipsés.